



ChangeViT: Unleashing plain vision transformers for change detection in remote sensing images

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ABSTRACT

Change detection in remote sensing images is essential for monitoring environmental changes on Earth's surface. Although vision transformers (ViTs), particularly plain ViTs pre-trained on large-scale datasets and endowed with extensive knowledge and strong comprehension capabilities, have achieved remarkable success as backbones in numerous computer vision applications, they remain underutilized in change detection tasks. In this domain, convolutional neural networks (CNNs) continue to dominate due to their powerful feature extraction capabilities. In this paper, our study uncovers ViTs' unique advantage in discerning large-scale changes, a capability where CNNs fall short. Capitalizing on this insight, we introduce ChangeViT, a framework that adopts a plain ViT backbone to enhance the performance of large-scale changes. This framework is supplemented by a detail-capture module that generates detailed spatial features and a feature injector that efficiently integrates fine-grained spatial information into high-level semantic learning. The feature integration ensures that ChangeViT excels in both detecting large-scale changes and capturing fine-grained details, providing comprehensive change detection across diverse scales. Without bells and whistles, ChangeViT achieves state-of-the-art performance on three popular high-resolution datasets (*i.e.*, LEVIR-CD, WHU-CD, and CLCD) and one low-resolution dataset (*i.e.*, OSCD), which underscores the unleashed potential of plain ViTs for change detection. Furthermore, thorough quantitative and qualitative analyses validate the efficacy of the introduced modules, solidifying the effectiveness of our approach. The source code is available at <https://github.com/zhuduowang/ChangeViT>.

1. Introduction

Change detection plays a crucial role in the field of remote sensing, employing pairs of bi-temporal images taken of the same geographic area at different times to track changes on the Earth's surface over time [1]. It has been widely applied in various applications such as disaster assessment [2], urban planning [3], and environmental management [4]. In recent years, convolutional neural networks (CNNs) have emerged as the primary backbone choice for state-of-the-art change detectors [2,5–9], as they can extract rich hierarchical features for detecting changes with different sizes.

Over the past few years, Vision Transformers (ViTs) [10] have *de facto* substituted CNNs as the dominant backbones in various computer vision tasks, *e.g.*, object detection [11,12] and image segmentation [13], which exhibit superior performance than CNN-based methods benefiting from the long-range modeling capability. While transformers have been

explored in the context of change detection in some preliminary studies [5,6,14,15], their performance has not yet matched that of the leading CNN models. Therefore, this paper aims to study the potential benefits of ViTs for change detection, striving to unleash their effectiveness in this area.

To assess the efficacy of ViTs in the change detection task, we first conduct a comprehensive performance comparison between change detectors utilizing ViTs and six established CNN architectures as backbones. The CNN backbones include three widely used models, *i.e.*, ResNet18 [16], VGG16 [17], and UNet [18], as well as three more recent advanced models, *i.e.*, MobileNetv3 [19], EfficientNetv2 [20], and ConvNeXtv2 [21]. This evaluation spans three well-known datasets, *i.e.*, LEVIR-CD [22], WHU-CD [23], and CLCD [24], as depicted in Fig. 1(a). Additionally, we explore the influence of various model initializations by incorporating pre-trained weights from DeiT [25], DINO [26], and DINOv2 [27] into our analysis. Specifically, ResNet18, VGG16, UNet,

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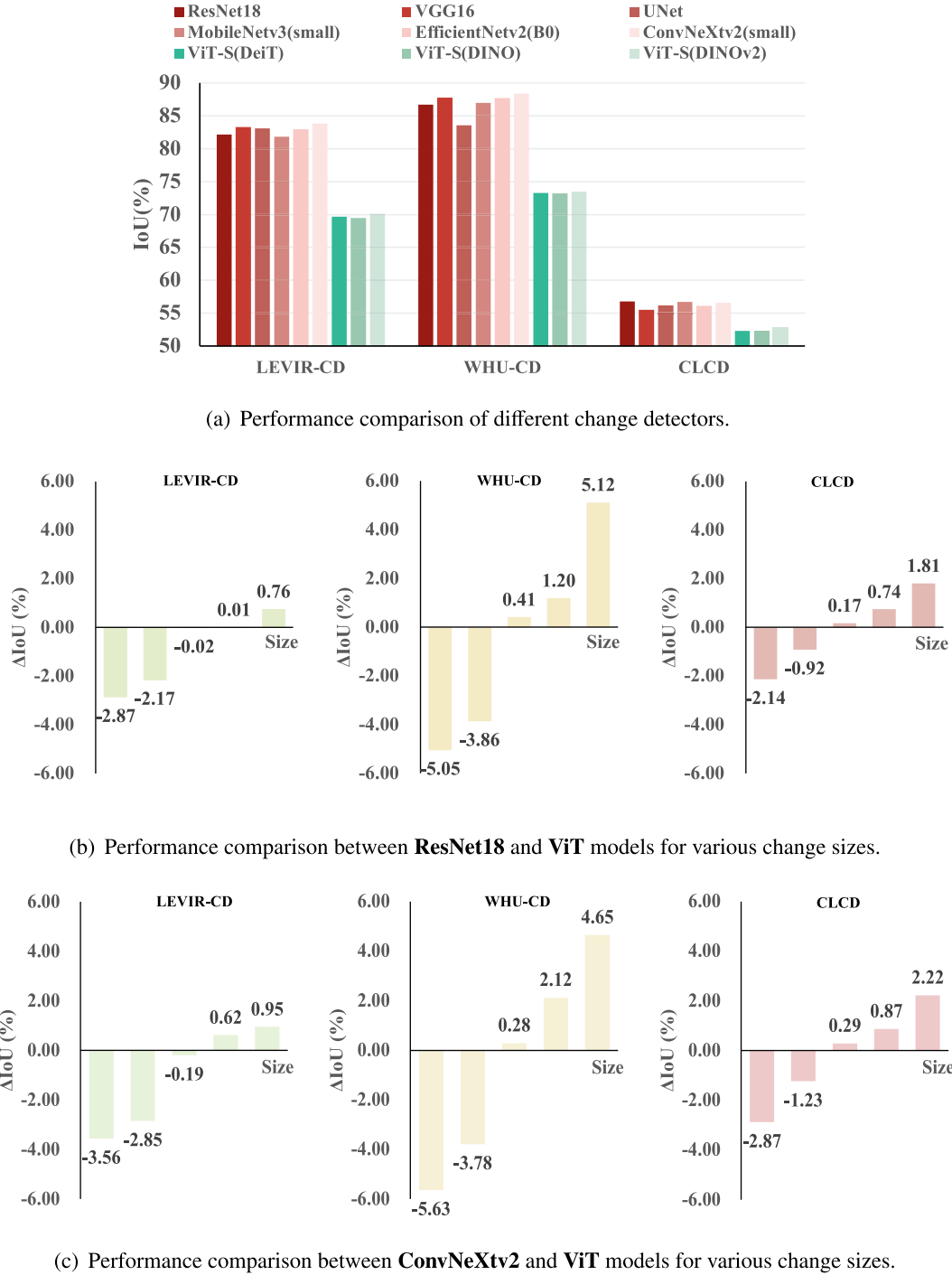


Fig. 1. (a) Performance comparison of different change detectors across three datasets, categorized as CNN-based and ViT-based models. (b)-(c) Performance comparison (ΔIoU (%)) between a typical CNN model (ResNet18), a state-of-the-art CNN model (ConvNeXtv2 (small)), and a ViT model (ViT-S (DINOv2)) for detecting changes with various sizes. The horizontal axis incrementally reflects the change sizes, progressing from smallest to largest changes. The ΔIoU values presented are calculated by subtracting the CNN's performance from that of the ViT for each size category.

MobileNetv3, EfficientNetv2, ConvNeXtv2, and ViT-S (DeiT) are pre-trained on ImageNet-1k with supervised training, while ViT-S(DINO) and ViT-S(DINOv2) are pre-trained with self-supervised training on ImageNet-1k, ImageNet-22k, and Google Landmarks, etc. The results indicate that: (1) CNN models significantly outperform all ViTs across all datasets, regardless of whether supervised or self-supervised learning is used, highlighting the dominance of CNNs in change detection tasks. (2) Even with identical data initialization (i.e., ImageNet-1k), the performance of ViTs remains inferior to that of CNN-based models.

To delve deeper into the models' capabilities, we perform an in-depth analysis of ViT (ViT-S with DeiT pre-training) and CNN models (ResNet18 and ConvNeXtv2 with ImageNet-1k pre-training) in detecting changes across various object sizes, which is illustrated in Fig. 1(b) and Fig. 1(c). We organize the test samples from each dataset by the proportion of pixels occupied by different objects within the images. Specifically, we first sort the images in ascending order based on the ratio of pixels occupied by changing objects to the total number of pixels in the image. Then, we evenly divide this ordered sequence into five

categories, ranging from the smallest to the largest proportions. We calculate the average performance difference between the ViT and CNN models within each category. The results show that though ViTs lag behind CNNs in detecting smaller changes, they demonstrate enhanced reliability for larger objects across all datasets. These insights suggest that while ViTs cannot capture fine-grained details as effectively as CNNs, they excel in detecting large-scale changes. Therefore, this previously untapped benefit has the potential to effectively mitigate the limitations inherent in CNN architectures.

Building upon the insights gathered from our preceding analysis, we propose **ChangeViT**, a simple yet effective framework that leverages the plain ViT framework as its core to capture large-scale object information. This is coupled with a detail-capture module specifically used to focus on fine-grained features. The detail-capture module functions as an auxiliary network, incorporating selected layers (C2-C4) from ResNet18 [16], which offers a more compact footprint (2.7M parameters) compared to a complete CNN model (11.2M parameters). To seamlessly inject these fine-grained details into the feature representation of ViTs, we establish connections between ViT's representations and fine-grained features. This integration is accomplished by considering ViT features as queries and merging the fine-grained features by applying the cross-attention mechanism.

Through extensive experiments on four widely recognized datasets, *i.e.*, LEVIR-CD [22], WHU-CD [23], CLCD [24], and OSCD [28], ChangeViT achieves the state-of-the-art performance across the board. In addition, we combine the proposed modules with various hierarchical transformers, *i.e.*, Swin Transformer [29], PVT [30], and PiT [31]. Consistently across these architectures, the proposed modules enhance performance, thereby further confirming their efficacy. Notably, despite the plain ViT's perceived limitations compared to these advanced hierarchical networks, ChangeViT outperforms methods that utilize these complex models, showcasing that we effectively unleash the capacity of plain ViTs in the field of change detection.

The main contributions of this paper can be summarized as follows:

- We thoroughly investigate the performance of plain ViTs and identify their aptitude for detecting large-scale changes. Motivated by this finding, we introduce ChangeViT, a simple yet effective framework that utilizes plain ViT as the primary feature extractor for the change detection task.
- To enhance the detection of changes across various sizes, we integrate a detail-capture module, specifically introduced to address the limitations of ViTs when identifying small objects. Furthermore, we introduce a feature injector to merge the extracted detailed features into high-level ones from the ViT, ensuring comprehensive feature representation within the model.
- ChangeViT achieves state-of-the-art performance on four popular datasets, *i.e.*, LEVIR-CD, WHU-CD, CLCD, and OSCD, demonstrating the superiority of the proposed method. Moreover, thorough quantitative and qualitative analyses validate the efficacy of the modules we have introduced, further solidifying the effectiveness of our approach.

2. Related work

2.1. Change detection

Existing deep learning-based change detection methods can be broadly classified into CNN-based and transformer-based architectures.

CNN-based Methods. CNN-based change detection approaches have been the mainstream framework in the literature [1,8,28,32–35] for a long time, known for their hierarchical feature modeling capabilities. These works primarily focus on multi-scale feature extraction, difference modeling, lightweight architecture designing, and foreground-background class imbalance. For instance, the method in [28] utilizes fully convolutional networks to capture hierarchical features for learn-

ing multi-scale feature representations. For adequate differential feature modeling, approaches in [32,33] incorporate the attention mechanism to establish relational dependencies among bi-temporal features. Method in [8] focuses on designing efficient and effective network architectures, utilizing lightweight feature extractors as backbones. Method in [1] addresses the significant challenge posed by foreground-background class imbalance by developing innovative loss functions that prioritize foreground alterations while minimizing interference from background noise (*e.g.*, seasonal variations, climate changes).

Transformer-based Methods. Recently, Vision Transformer (ViT) [10] and its variants [29,30] have surpassed CNN in various visual tasks and become the dominant backbone [11,12,36]. Motivated by these achievements, several works [5,6,14,15,37,38] have explored the application of transformers in change detection tasks. Some of these methods [14,38] utilize pure transformers, while others [5,6,37,39] adopt CNN-Transformer hybrid architectures. Methods in [14,38,40] introduce hierarchical transformer networks based on the swin transformer [29]. The others typically follow a paradigm in which features extracted by CNN serve as semantic tokens, followed by contextual relation modeling between bi-temporal tokens using transformer blocks. The method introduced in [15] exhibits an efficient tuning strategy that involves freezing the parameters of the Transformer encoder while introducing additional trainable parameters. However, this method fails to deliver optimal results due to an inadequate exploration of the strengths and limitations of the transformers. This precludes a more effective application of the model's capabilities, thereby capping the potential gains in performance.

Building upon these transformer-based methods, ChangeViT distinguishes itself by adopting a pure ViT architecture as the cornerstone feature extractor, diverging from traditional hierarchical CNN or Transformer-based networks. This pure ViT backbone facilitates superior semantic extraction and reveals previously unidentified potential in detecting large-scale changes. Additionally, ChangeViT leverages extensive pre-training and incorporates specialized modules for fine-grained feature integration. Therefore, ChangeViT effectively integrates high-level semantic information with detailed spatial features, offering a novel approach within the change detection field.

2.2. Plain ViT for downstream tasks

ViT [10] is a plain, non-hierarchical architecture, which is a powerful alternative to standard CNN for image classification. Due to the significant computational overhead of self-attention in ViT, subsequent works focus on designing more efficient architectures, such as Swin [29], PVT [30], PiT [31], and AnchorFormer [41]. These works inherit some designs from CNN, including hierarchical structures, sliding windows, and convolutions. Recently, researchers have begun to study the potential of ViT for various downstream tasks motivated by the emergence of large pre-trained models, *e.g.*, DeiT [25], DINO [26], and DINOv2 [27]. The plain ViT has already made remarkable progress in dense prediction [36]. ViT-Adapter [36] introduces a pre-training-free adapter that injects prior knowledge to ViT without redesigning its architecture for various dense prediction tasks. This trend is further exemplified in the remote sensing domain. Models such as RingMoE [11], RingMo-Aerial [12], and SkySense [13] have demonstrated superior performance across more than 20 tasks, all without requiring architectural redesign.

Inspired by the above works, we aim to unleash the potential of the plain ViT model, enabling it to adapt well to change detection tasks.

3. Proposed method

The overall architecture of the proposed method, depicted in Fig. 2, integrates high-level semantic understanding with fine-grained detail extraction to address the challenges of change detection in bi-temporal

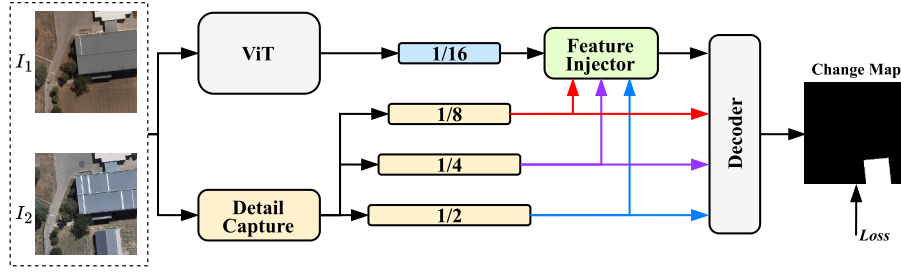


Fig. 2. Overview of the proposed **ChangeViT**. Bi-temporal images I_1 and I_2 are first fed into a shared ViT to extract high-level semantic features and a detail-capture module to extract low-level detailed information. Subsequently, a feature injector is introduced to inject the low-level details into high-level features. Finally, a decoder is utilized to predict probability maps.

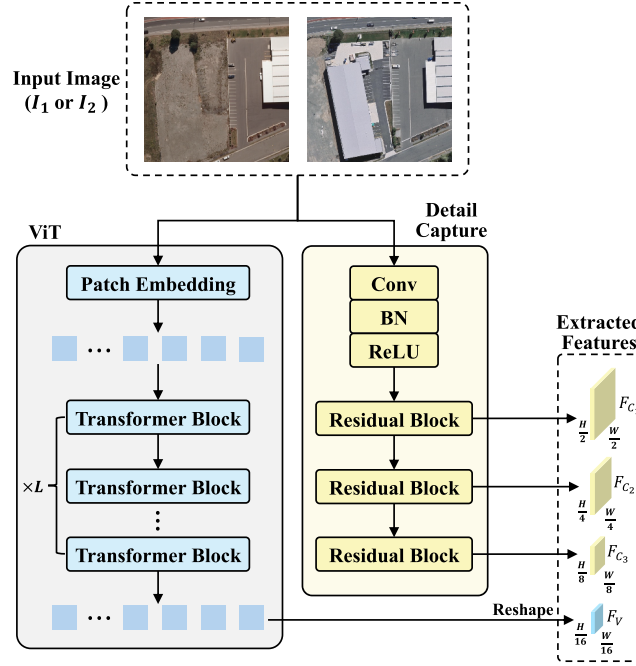


Fig. 3. Detailed architecture of the feature extractor. It comprises a plain ViT with L Transformer blocks and a detail-capture module made up of three residual blocks. This design produces multi-scale detailed features and high-level semantic features, both essential for effective change detection.

images. Given input images $I_1 \in \mathbb{R}^{H \times W \times 3}$ and $I_2 \in \mathbb{R}^{H \times W \times 3}$, the framework processes them in parallel through a Vision Transformer (ViT) and a detail-capture module. The ViT extracts high-level semantic features, $F_V^t \in \mathbb{R}^{\frac{H}{16} \times \frac{W}{16} \times C_4}$ ($t \in \{1, 2\}$), which encapsulate global semantic patterns crucial for identifying large-scale changes. Meanwhile, the detail-capture module focuses on fine-grained local features, producing multi-scale feature maps, $F_{C_i}^t \in \mathbb{R}^{\frac{H}{2^i} \times \frac{W}{2^i} \times C_i}$ ($i \in \{1, 2, 3\}$), at resolutions of $\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}$ to the original image size. These fine-grained features are critical for detecting subtle or localized changes that may not be captured by the ViT alone.

To facilitate effective interaction between the semantic and detailed features, a feature injector is employed to integrate the low-level features from the detail-capture module with the high-level features from the ViT. This integration enhances the semantic representations by preserving intricate spatial details within the high-level feature space. Finally, a multi-scale feature fusion decoder combines and refines these enriched features to produce the final change probability map, $M \in \mathbb{R}^{H \times W}$. By progressively aggregating information across scales, the decoder restores the resolution of the change map to its original size while maintaining precision in detecting changes of varying scales and complexities.

3.1. Feature extractor

As illustrated in Fig. 3, the feature extractor comprises two primary components: a plain Vision Transformer (ViT) and a detail-capture module. These components are detailed below:

Plain ViT. The plain ViT serves as the backbone for extracting high-level semantic features from bi-temporal images. Given input images $I_1, I_2 \in \mathbb{R}^{H \times W \times 3}$, they are first processed by a patch embedding layer. This layer divides each image into non-overlapping patches of size 16×16 , resulting in $N = \frac{H \times W}{256}$ patches per image. Each patch is flattened and transformed into a C_4 -dimensional token vector through a learnable projection:

$$T_p^t = \text{Flatten}(\text{PatchEmbed}(I_t)), \quad (1)$$

where $T_p^t \in \mathbb{R}^{N \times C_4}$ ($t \in \{1, 2\}$) represents the embedded tokens, and $\text{PatchEmbed}(\cdot)$ denotes a 2D convolution with a kernel size and stride of 16×16 .

To preserve spatial positional information, a learnable positional embedding $P \in \mathbb{R}^{N \times C_4}$ is added to the patch embeddings:

$$T_0^t = T_p^t + P. \quad (2)$$

The tokens, T_0^t , are then passed through L stacked Transformer blocks, where each block comprises a Multi-Head Self-Attention (MHSA) mechanism, a Feed-Forward Network (FFN), and Layer Normalization (LN). The operations in each block are defined as follows:

$$T_l^t = \text{LN}(T_{l-1}^t) + \text{MHSA}(\text{LN}(T_{l-1}^t)), \quad (3)$$

$$T_l^t = T_l^t + \text{FFN}(\text{LN}(T_l^t)), \quad (4)$$

where $l \in \{1, \dots, L\}$ denotes the block index, and $\text{FFN}(\cdot)$ comprises two fully connected layers interleaved with a GELU activation function.

Finally, the output of the last Transformer block is reshaped back to the spatial dimensions. The resulting high-level feature map for each image is expressed as:

$$F_V^t = \text{Reshape}(T_L^t), \quad (5)$$

where $F_V^t \in \mathbb{R}^{\frac{H}{16} \times \frac{W}{16} \times C_4}$ represents the high-level semantic representations which are critical for detecting large-scale changes.

Detail-Capture. As discussed in Section 1, while the plain ViT is effective at capturing global semantic features, it struggles with detecting subtle, fine-grained local changes. To address this limitation, we introduce a detail-capture module designed to extract localized features at multiple scales.

The detail-capture module consists of three residual convolutional blocks (C2-C4) adapted from ResNet18 [16]. These blocks progressively downsample the input images, extracting multi-scale detailed features. Initially, the input image I_t is passed through the convolutional block and the first residual block to produce the feature map $F_{C_1}^t$. This is followed by two residual blocks, which generate the feature maps $F_{C_2}^t$ and $F_{C_3}^t$, respectively. The resolutions of these feature maps correspond to the scales $\{\frac{1}{2}, \frac{1}{4}, \frac{1}{8}\}$ of the original image size. These multi-scale feature maps, $F_{C_i}^t \in \mathbb{R}^{\frac{H}{2^i} \times \frac{W}{2^i} \times C_i}$, with $i \in \{1, 2, 3\}$, are crucial for capturing fine-grained details that are important for detecting small or localized changes.

By integrating the high-level semantic features extracted by the ViT with the multi-scale fine-grained features generated by the detail-capture module, the proposed feature extractor effectively captures comprehensive information, enabling the detection of diverse changes.

3.2. Feature injector

In the context of change detection, preserving detailed spatial features is crucial for effectively identifying subtle changes and small objects. These features are typically located in the lower layers of the network and tend to diminish as they propagate to deeper layers. Thus, ensuring an efficient transmission mechanism for integrating low-level spatial details into high-level semantic representations is critical for enhancing detection performance.

To address this challenge, we propose the feature injector, a module designed to bridge the gap between low-level and high-level features. The feature injector comprises three cross-attention blocks [42], as illustrated in Fig. 4(a) and (c). The low-level detailed features are used as keys (K) and values (V), whereas the high-level semantic features extracted by the ViT serve as queries (Q). This configuration is both intuitive and effective, enabling the feature injector to selectively incorporate relevant spatial details into the semantic representation based on query information. The operation of the cross-attention is formulated as follows:

$$F_{V_E}^{t,i} = \text{CrossAttn}(F_V^t, F_{C_i}^t), \quad (6)$$

where F_V^t represents the high-level semantic features, and $F_{C_i}^t$ denotes the multi-scale low-level features. This formulation allows the module to aggregate detailed information from various scales, refining the semantic feature representation and enriching its spatial context.

To further consolidate the information gathered from various scales, the outputs from the three cross-attention blocks are concatenated along the channel dimension and then passed through a 1×1 depth-wise convolution to produce the enhanced high-level representation:

$$F_{V_E}^t = \text{FC}(F_{V_E}^{t,1} \odot F_{V_E}^{t,2} \odot F_{V_E}^{t,3}). \quad (7)$$

Here, $\text{FC}(\cdot)$ represents a 2D depth-wise convolution with a kernel size of 1×1 , and $\odot$ denotes the concatenation operation along the channel dimension. The resulting enhanced features, $F_{V_E}^t$, incorporate fine-grained details while retaining global semantic understanding, thereby making them more robust for detecting changes at varying scales.

In addition to the standard feature injector, we explore an alternative design, illustrated in Fig. 4(b). Unlike the initial design, this approach reverses the roles of low-level and high-level features. Specifically, low-level features act as queries, while high-level semantic features from the ViT serve as keys and values. This configuration allows the feature injector to refine hierarchical detailed features by leveraging the semantic context encoded in high-level features.

This refinement mechanism ensures that low-level features are adjusted in accordance with global semantics, further enhancing their ability to represent intricate spatial details and complement high-level representations. We analyze the comparative performance of these two designs in subsequent experiments to assess their respective contributions to the overall feature extraction process.

3.3. Decoder and optimization

In contrast to existing methods [1,6,33,37], which employ complex architectures and sophisticated techniques to model differential information and predict the change probability map, we adopt a simpler yet effective decoder design in ChangeViT. This choice underscores our intention to highlight the learning capabilities of the proposed ChangeViT framework without the confounding effects of overly intricate decoder mechanisms.

As illustrated in Fig. 5, we use a straightforward feature fusion layer to capture differences between bi-temporal features. A cascade convolutional layer, followed by an upsampling operation, is employed to progressively aggregate differential features from deep to shallow layers, ultimately restoring them to the original resolution of $H \times W$. The difference modeling is formulated as Eq. (8):

$$F_{D_i} = \text{MLP}(F_i^1 \odot F_i^2 \odot |F_i^1 - F_i^2|), \quad (8)$$

where $F_i^t \in \{F_{C_1}^t, F_{C_2}^t, F_{C_3}^t, F_{V_E}^t\}$ ($t \in \{1, 2\}$), MLP is a three-layer 2D convolutional network with kernel size of 3×3 along with ReLU activation function, $\odot$ denotes concatenating on channel dimension, and $|\cdot|$ means absolute value operation.

The concatenation of F_i^1 and F_i^2 captures the combined spatial and semantic information from both temporal inputs. The inclusion of the absolute difference $|F_i^1 - F_i^2|$ highlights the regions where changes have occurred, thereby enhancing the discriminative power of the feature representations for subsequent processing.

To progressively aggregate differential features from deep to shallow layers and restore them to the original resolution of $H \times W$, we employ a cascade of convolutional layers followed by upsampling operations. This process is formulated as:

$$F_{D_i} \leftarrow \text{Deconv}_{4 \times 4}(\text{Conv}_{1 \times 1}(F_{D_{i+1}})) + F_{D_i}, \quad (9)$$

where $\text{Conv}_{1 \times 1}$ is a 2D convolution with a kernel size of 1×1 to reduce the channel dimension, and $\text{Deconv}_{4 \times 4}$ denotes 2D deconvolution to up-sample the feature map with a kernel size of 4×4 and a stride size of 2×2 . The addition $+ F_{D_i}$ incorporates residual connections, which help in preserving spatial details and mitigating vanishing gradient issues during training.

This cascading approach allows the decoder to methodically reconstruct the high-resolution change map by aggregating multi-scale differential features. Starting at the deepest layer where the most abstract

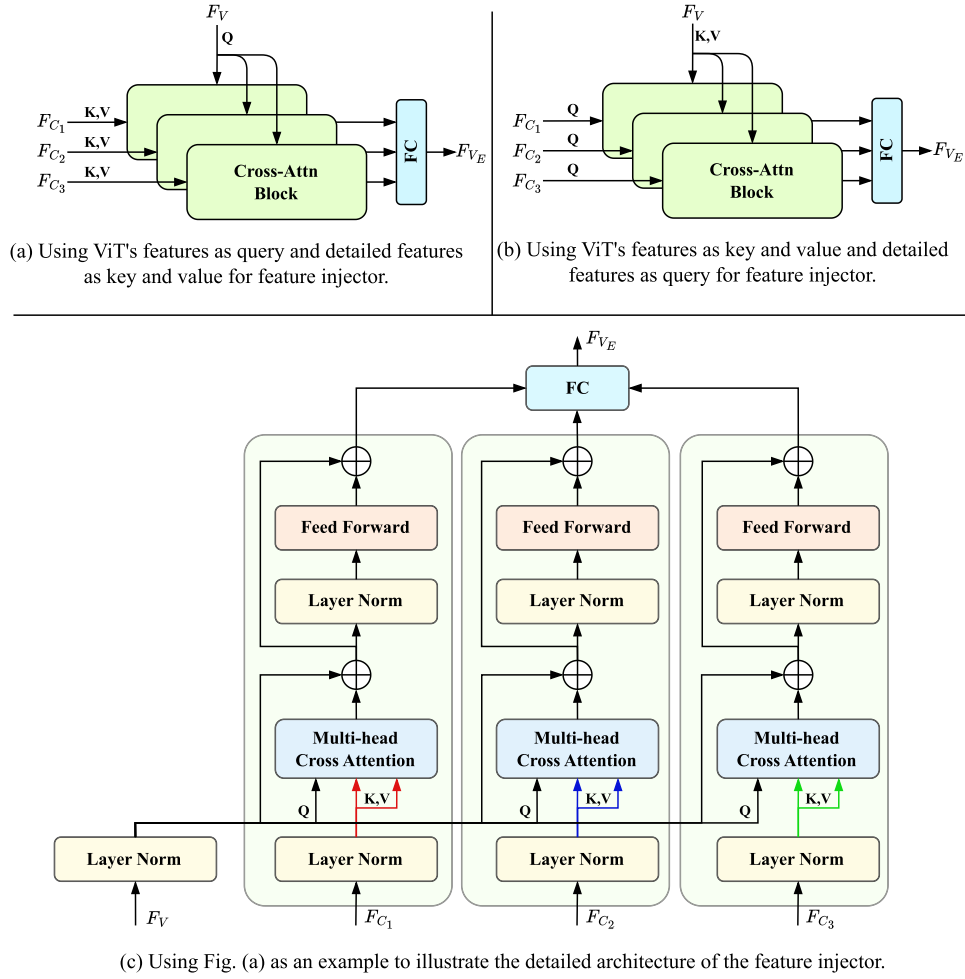


Fig. 4. Overview of the feature injectors. F_{C_i} ($i \in \{1, 2, 3\}$) are multi-scale detailed features, and F_V is the ViT feature. (a) F_V as query, F_{C_i} as key/value to inject details into ViT. (b) F_{C_i} as query, F_V as key/value to refine ViT features. (c) Detailed feature injector architecture.

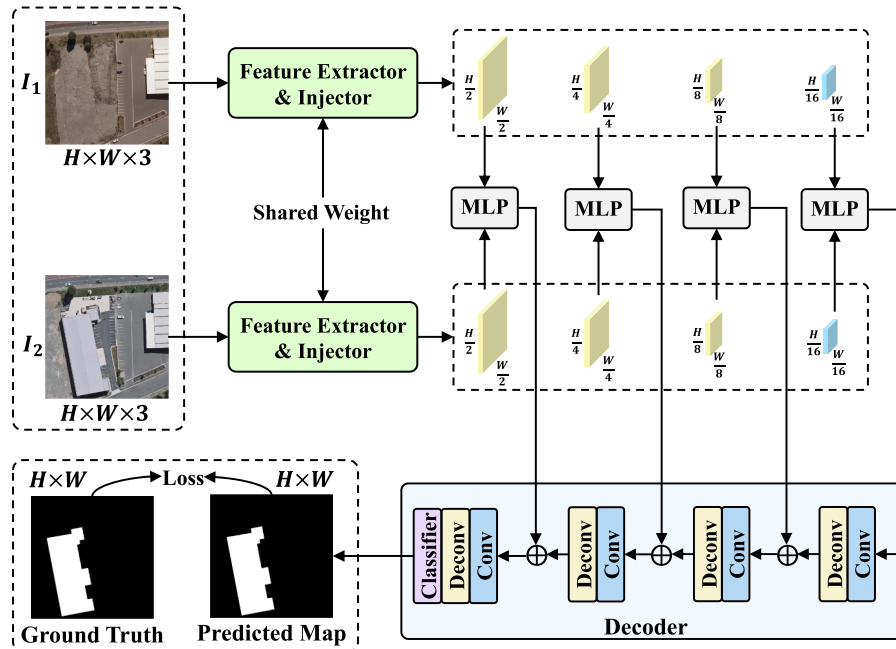


Fig. 5. Overview of the detailed architecture of ChangeViT. A shared-weight feature extractor and injector process bi-temporal images to generate bi-temporal feature representations. Then, several MLPs are applied to obtain differential features, which are finally processed by a simple cascade decoder to predict the changes.

representations are formed, each upsampling step improves the feature maps by integrating finer details from shallower layers, ensuring that the final output maintains both wide-ranging contextual information and precise feature details.

Finally, a classification layer is applied to transform the shallowest features F_{D_1} into change maps M , which is formulated as Eq. (10):

$$M = \text{Sigmoid}(\text{Conv}_{3 \times 3}(F_{D_1})). \quad (10)$$

where $\text{Conv}_{3 \times 3}$ is a 2D convolution with the kernel size of 3×3 , Sigmoid function maps the feature map to $(0, 1)$ and then transforms to a binary map given a predefined threshold (*i.e.*, 0.5), *i.e.*, $M \in \{0, 1\}^{H \times W}$.

As mentioned in prior works [1,8], the proportion of changed targets is significantly lower than that of unchanged ones. Following the above works, we adopt binary cross-entropy (BCE) and dice loss (Dice) [1] to alleviate the class imbalance problem. The change detection loss $\mathcal{L}_{total}$ is defined as Eq. (11):

$$\mathcal{L}_{total} = \mathcal{L}_{bce}(P, Y) + \mathcal{L}_{dice}(P, Y), \quad (11)$$

The BCE and Dice losses are formulated as follows:

$$\begin{aligned} \mathcal{L}_{bce}(P, Y) &= -\frac{1}{N} \sum_{i=1}^N [Y_i \log_2 P_i + (1 - Y_i) \log_2 (1 - P_i)], \\ \mathcal{L}_{dice}(P, Y) &= 1 - \frac{2 \sum_{i=1}^N P_i Y_i + \epsilon}{\sum_{i=1}^N (P_i)^2 + \sum_{i=1}^N (Y_i)^2 + \epsilon}, \end{aligned} \quad (12)$$

where i denotes the i -th pixel, N is the number of total pixels, Y denotes the ground truth and ϵ (*i.e.*, $1e-5$) is a smooth term utilized to avoid zero division.

4. Experiments

We conducted extensive experiments on three widely used high-resolution datasets, namely LEVIR-CD [22], WHU-CD [23], and CLCD [24], as well as one challenging low-resolution dataset, OSCD [28], to demonstrate the effectiveness of the proposed method. To better understand each component of ChangeViT, we conduct extensive diagnostic experiments in Section 4.5. Otherwise stated, we use ChangeViT-S for experiments on the three high-resolution datasets.

4.1. Implementation details

We adopt vanilla ViT [10] as our primary backbone, specifically incorporating its tiny and small variants, thereby constructing two models named ChangeViT-T and ChangeViT-S. We use DeiT [25] and DINOv2 [27] pre-trained weights for initialization, respectively. Our models are implemented using the PyTorch framework and executed on a computing platform consisting of a single NVIDIA GeForce RTX 3090 GPU paired with an Intel(R) Xeon(R) Gold 6138 CPU. For optimization, we opt for the Adam optimizer, with beta values set to (0.9, 0.99) and a weight decay of $1e-4$. Initially, the learning rate is $2e-4$ and gradually reduces according to a scheduled reduction formula: $(1 - (\text{curr_iter}/\text{max_iter}))^\alpha \times \text{lr}$, where α is set to 0.9 and max_iter is set to 80K iterations on LEVIR-CD and WHU-CD, 40K for CLCD dataset, and 10K for OSCD, respectively. The batch size remains at 16 across all experiments. To augment the training data and bolster the model's robustness, we apply random flipping and cropping data enhancement approaches. The channel dimensions of F_{C_i} are set to 64, 128, and 256, respectively. Furthermore, we ensure consistency and fairness in comparison by meticulously aligning the experimental settings of the compared methods with those specified in the original paper.

4.2. Datasets

LEVIR-CD [22] contains 637 pairs of high-resolution (1024×1024 , 0.5 m/pixel) bi-temporal images from various cities in Texas, collected between 2002 and 2018. Each image is cropped into 16 patches of

256×256 , resulting in 7120 training, 1024 validation, and 2048 test pairs.

WHU-CD [23] is a building change detection dataset with high-resolution (0.2m) aerial images (32507×15354 pixels), mainly covering post-earthquake reconstruction areas. Images are split into 256×256 patches, yielding 5947 training, 744 validation, and 744 test pairs.

CLCD [24] comprises bi-temporal images (0.5-2m resolution) of Guangdong, China, acquired in 2017 and 2019, focusing on cropland and land cover changes. After cropping to 256×256 patches, the dataset provides 1440 training, 480 validation, and 480 test pairs.

OSCD [28] includes Sentinel-2 images (10-60 m resolution) from 24 global urban regions (600×600 pixels), mainly for large-scale urban change detection. Images are divided into 256×256 patches, with 75 training and 28 test pairs.

4.3. Evaluation metrics

Following the widely used evaluation protocols in the change detection task, we use three accuracy metrics, *i.e.*, F1 score (F1), intersection over union (IoU), and overall accuracy (OA), to evaluate our proposed method. They are formulated as follows:

$$\begin{aligned} P &= \frac{TP}{TP + FP}, \quad R = \frac{TP}{TP + FN}, \quad F1 = \frac{2PR}{P + R}, \\ IoU &= \frac{TP}{TP + FN + FP}, \quad OA = \frac{TP + TN}{TP + TN + FN + FP}, \end{aligned} \quad (13)$$

where TP, FP, FN, and TN indicate true positive, false positive, true negative, and false negative, respectively. For all the metrics, a higher value means better detection performance.

4.4. Comparison with state-of-the-art approaches

As illustrated in Table 1, we compare ChangeViT with the previous methods on three high-resolution datasets, *i.e.*, LEVIR-CD, WHU-CD, and CLCD. Notably, all the compared methods employed a hierarchical backbone as primary feature extractors. Specifically, DTCDSN, BIT, ICIFNet, DMINet, GASNet and AMTNet apply ResNet [16] or its variants [44] as backbones, SNUNet and EATDer apply nested UNet [18] and stack non-local blocks [45], respectively. In contrast, our approach employs a non-hierarchical, plain ViT as the core backbone, which includes ViT-T and ViT-S, coupled with a lightweight detail-capture module, which serves as an auxiliary network. From Table 1, we can summarize the following valuable findings: (1) ChangeViT consistently outperforms the existing works across all datasets and evaluation metrics, despite utilizing the tiny backbone of ViT, which demonstrates its effectiveness. (2) The primary feature extractor in ViTs, despite being non-hierarchical, demonstrates competitive performance when compared to hierarchical-based methods. This underscores the robust feature extraction and representation capabilities that large-scale pre-training ViT can offer, fully realizing its potential. (3) Notably, ChangeViT-T and ChangeViT-S exhibit significant performance gains over the SOTA method (*i.e.*, AMTNet) by 3.99% and 4.54% IoU on the WHU-CD dataset. This finding is sensible given that the changes in WHU-CD vary widely, with fewer medium-sized objects compared to smaller and larger ones. This observation aligns with the results illustrated in the middle of Fig. 1(b), underscoring the efficacy of our proposed method in capturing global features and extracting fine-grained spatial information. (4) With an increase in the size of the primary feature extractor, ChangeViT demonstrates enhanced performance. Notably, the detail-capture module, comprising just 2.7M parameters, stands out for its lightweight nature when compared to the total parameter count of each model (11.68M and 32.13M). Our proposed ChangeViT achieves a superior balance between efficiency and effectiveness compared to previous methods, underscoring its superiority.

As shown in Table 1, we also compare ChangeViT with several existing methods on the low-resolution dataset, *i.e.*, OSCD. The targets in

Table 1

Performance comparison of different change detection methods on LEVIR-CD, WHU-CD, CLCD, and OSCD datasets, respectively. The best results are highlighted in **bold** and the second-best results are underlined. All results of the three evaluation metrics are described as percentages (%).

Method	#Params(M)	FLOPs(G)	LEVIR-CD			WHU-CD			CLCD			OSCD		
			F1	IoU	OA	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
DTCDSN [34]	41.07	20.44	87.43	77.67	98.75	79.92	66.56	98.05	57.47	40.81	94.59	36.13	22.05	94.50
SNUNet [43]	12.04	54.82	88.16	78.83	98.82	83.22	71.26	98.44	60.82	43.63	94.90	27.02	15.62	93.81
ChangeFormer [14]	41.03	202.79	90.40	82.48	99.04	87.39	77.61	99.11	61.31	44.29	94.98	38.22	23.62	94.53
BIT [5]	3.55	10.63	89.31	80.68	98.92	83.98	72.39	98.52	59.93	42.12	94.77	29.58	17.36	90.15
ICIFNet [32]	23.82	25.36	89.96	81.75	98.99	88.32	79.24	98.96	68.66	52.27	95.77	23.03	13.02	94.61
DMINet [33]	6.24	<u>14.42</u>	90.71	82.99	99.07	88.69	79.68	98.97	67.24	50.65	95.21	42.23	26.76	95.00
GASNet [1]	23.59	23.52	90.52	83.48	99.07	91.75	84.76	99.34	63.84	46.89	94.01	10.71	5.66	91.52
AMTNet [6]	24.67	21.56	90.76	83.08	98.96	92.27	85.64	99.32	75.10	60.13	96.45	10.25	5.40	94.29
EATDer [37]	6.61	23.43	91.20	83.80	98.75	90.01	81.97	98.58	72.01	56.19	96.11	54.23	36.98	93.85
ChangeViT-T	11.68	27.15	<u>91.81</u>	<u>84.86</u>	<u>99.17</u>	<u>94.53</u>	<u>89.63</u>	<u>99.57</u>	<u>77.31</u>	<u>63.01</u>	<u>96.67</u>	<u>55.13</u>	<u>38.06</u>	<u>95.01</u>
ChangeViT-S	32.13	38.80	91.98	85.16	99.19	94.84	90.18	99.59	77.57	63.36	96.79	55.51	38.42	95.05

Table 2

Study the effectiveness of proposed modules with different transformer architectures on three datasets, respectively. The check mark (✓) denotes combining with our proposed modules. All results are described as percentages (%).

Architecture	Model	Ours	LEVIR-CD			WHU-CD			CLCD		
			F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
Hierarchical	Swin-S [29]	✓	89.40	80.83	98.94	93.03	86.98	99.22	73.80	58.47	96.33
			90.18	82.11	99.01	94.04	88.75	99.30	75.41	60.52	96.40
	PVT-S [30]	✓	84.60	73.31	98.38	87.36	77.55	98.89	70.25	54.15	95.76
			87.26	77.40	98.68	89.09	80.32	98.92	71.95	56.19	95.90
	PiT-S [31]	✓	84.94	73.83	98.38	87.34	77.53	98.89	70.01	53.86	95.88
Non-hierarchical	TNT-S [46]	✓	87.20	77.31	98.63	89.50	81.00	98.94	72.80	57.23	95.93
			82.42	71.10	98.01	84.20	73.82	98.51	69.95	52.82	95.65
	GPViT-L1 [47]	✓	91.82	84.87	99.06	94.56	89.68	99.43	77.10	63.08	96.32
			82.58	70.46	98.32	84.25	73.12	98.58	68.89	52.34	95.51
	ViT-S [27]	✓	91.86	85.05	99.15	94.55	90.02	99.40	77.41	63.10	96.53
			82.39	70.05	98.25	84.70	73.46	98.82	69.05	52.74	95.75
			91.98	85.16	99.19	94.84	90.18	99.59	77.53	63.30	96.76

the OSCD are relatively smaller than those in high-resolution datasets, exacerbating the foreground-background imbalance issue and making it challenging to detect smaller targets. From Table 1, the following key points can be noted: (1) The proposed ChangeViT outperforms all compared methods across three evaluation metrics, despite utilizing tiny or small models of ViT, demonstrating its effectiveness on the low-resolution dataset. (2) GASNet and AMTNet perform poorly on this dataset, likely due to their inefficiency in detecting small targets. Although GASNet introduces a foreground-awareness module to address the category imbalance between the foreground and background, it still underperforms in detecting changes in low-resolution remote sensing images.

4.5. Diagnostic study

Effectiveness with different architectures. In Table 2, we investigate the effectiveness of the proposed modules with different architectures, including hierarchical (*i.e.*, Swin-S [29], PVT-S [30], PiT-S [31]) and non-hierarchical (*i.e.*, TNT-S [46], GPViT-L1 [47], and ViT-S [10]) transformers. Key observations from the table include: (1) Without combining with our proposed modules, the non-hierarchical ViT-S underperforms the other hierarchical methods across all metrics on three datasets. (2) When integrated with our proposed modules, all transformers exhibit performance improvements, indicating the efficacy of our approach regardless of transformer architectures, *i.e.*, hierarchical or non-hierarchical. (3) Among the compared architectures, ViT-S consistently demonstrates significant performance gains when equipped with our proposed modules, outperforming both hierarchical and other non-hierarchical transformers. This suggests that our modules effectively

mitigate ViT-S's limitations in capturing detailed information and accurately detecting changes across diverse scales.

Effectiveness of proposed modules. To investigate the effectiveness of the proposed modules, we conduct comprehensive diagnostic experiments on LEVIR-CD, WHU-CD, and CLCD datasets. Table 3 presents the performance gains introduced by each module across three datasets. Specifically, incorporating the detail-capture module results in significant improvements in IoU by **13.77 %**, **13.97 %**, and **7.58 %** for LEVIR-CD, WHU-CD, and CLCD, respectively, compared to the baseline ViT model. This clearly illustrates the detail-capture module's effectiveness in capturing fine-grained spatial details. Furthermore, integrating the feature injector leads to additional IoU enhancements of **1.34 %**, **2.75 %**, and **2.84 %** across three datasets compared to the model with only the detail-capture module. These improvements underscore the feature injector's capability to seamlessly merge detailed spatial information into high-level semantic features, thereby further boosting the overall performance of ChangeViT. In summary, all of our proposed modules are essential and effective in the ChangeViT framework.

Effectiveness of different CNNs within detail-capture. To investigate the effectiveness of the proposed detail-capture module with different CNNs, including ResNet18 and more lightweight models (*i.e.*, MobileNetv3 [48], EfficientNetv2 [20], and ShuffleNetv2 [49]), we conduct experiments as presented in Table 4. The results demonstrate that: (1) ChangeViT integrated with a detail-capture module, whether using lightweight CNNs or ResNet18, exhibits significant performance gains. This underscores the generality and effectiveness of our proposed ChangeViT framework. (2) ResNet18 layers, when used as the detail-capture module, achieve the most competitive performance

Table 3

Effectiveness of the proposed modules in ChangeViT on three datasets, respectively. DC and FI denote the detail-capture and feature injector, respectively. All results are presented as percentages (%). The gains from each module are shown beneath each metric compared to the baseline.

Module	LEVIR-CD			WHU-CD			CLCD		
	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
Baseline (ViT)	82.39	70.05	98.25	84.70	73.46	98.82	69.18	52.88	95.68
Baseline + DC	91.20 (+8.81)	83.82 (+13.77)	99.11 (+0.86)	93.30 (+8.60)	87.43 (+13.97)	99.46 (+0.64)	75.36 (+6.18)	60.46 (+7.58)	96.62 (+0.94)
Baseline + DC + FI	91.98 (+9.59)	85.16 (+15.11)	99.19 (+0.94)	94.84 (+10.14)	90.18 (+16.72)	99.59 (+0.77)	77.53 (+8.35)	63.30 (+10.42)	96.76 (+1.08)

Table 4

Study the effectiveness of the proposed detail-capture module with different CNNs, including ResNet18 and lightweight models (*i.e.*, MobileNetv3, EfficientNetv2, and ShuffleNetv2), on three datasets, respectively. DC denotes the detail-capture module.

Model	#Params(M)	FLOPs(G)	LEVIR-CD			WHU-CD			CLCD		
	(DC / Total)	(DC / Total)	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
-	- / 21.59	- / 13.94	82.39	70.05	98.25	84.70	73.46	98.82	69.18	52.88	95.68
MobileNetv3 (small)	0.93 / 28.25	0.63 / 16.42	91.34	84.38	98.97	94.22	88.96	99.33	76.52	61.66	96.50
EfficientNetv2 (B0)	0.04 / 27.44	0.28 / 16.05	91.22	84.30	98.93	94.03	88.75	99.18	76.85	62.16	96.52
ShuffleNetv2 (0.5 ×)	0.14 / 28.59	1.12 / 21.72	91.06	84.02	98.83	94.12	88.58	99.19	76.38	61.58	96.25
ResNet18	2.70 / 32.13	3.68 / 38.80	91.98	85.16	99.19	94.84	90.18	99.59	77.53	63.30	96.76

Table 5

Investigation on the impact of multiple scales in the detail-capture module on three datasets, respectively. All results are described as percentages (%).

Scales			LEVIR-CD			WHU-CD			CLCD		
1/8	1/4	1/2	F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
✓			91.32	84.03	99.12	94.20	89.04	99.55	76.43	61.85	96.49
	✓		91.08	83.62	99.10	92.90	86.74	99.45	73.25	57.79	96.31
		✓	89.43	80.89	98.94	90.87	83.28	99.29	70.82	54.82	95.79
✓	✓		91.56	84.43	99.15	94.25	89.07	99.58	77.30	63.07	96.62
✓		✓	91.45	84.24	99.14	94.02	88.67	99.44	77.10	62.73	96.56
	✓	✓	90.94	83.39	99.09	92.49	86.04	99.41	75.22	60.28	96.46
✓	✓	✓	91.98	85.16	99.19	94.84	90.18	99.59	77.53	63.30	96.76

compared to lightweight models. Although lightweight models have fewer parameters and lower FLOPs, they are insufficient for capturing the fine-grained details essential in remote sensing insufficient for capturing the fine-grained images.

Impact of multiple scales. To investigate the necessity of capturing multiple scales in the detail-capture module, we conduct experiments using multi-scale features, *i.e.*, 1/2, 1/4, 1/8. As shown in Table 5, we can get the following key observations: (1) Single-scale features often yield subpar results, while the amalgamation of multi-scale features leads to enhanced performance. (2) An interesting finding is that high-level features or their combinations can achieve better performance than low-level features. (3) Furthermore, the inclusion of three-scale features results in mutual improvements, indicating that multi-scale features leverage spatial cues across complementary levels.

Impact of pre-trained weights. To investigate the impact of pre-trained weights on ChangeViT, we apply various model initialization approaches, including random initialization and several publicly available large-scale pre-trained weights derived from both supervised and self-supervised training strategies on various datasets. As illustrated in Table 6, we observed the following key points: (1) Both ChangeViT-T and ChangeViT-S exhibit improved detection accuracy when utilizing pre-trained weights compared to random initialization. (2) DINOv2-S provides the most effective pre-trained weights for the ChangeViT-S model, benefiting from large-scale data pre-training. (3) When DMiNet, GASNet, AMTNet, and ChangeViT are pre-trained on the same data,

i.e., ImageNet-1k, the proposed ChangeViT outperforms all the CNN-based methods, demonstrating the effectiveness of transferring the priorities of large pre-trained ViT models to the change detection task.

Impact of different fusion approaches in feature injector. Several experiments are conducted to investigate the impact of different feature fusion approaches, including addition, concatenation, transformer decoder blocks, and feature injector, as shown in Table 7. (1) **Setup:** For **addition fusion**, we first resize all multi-scale low-level features $F_{C_i}^t$ to match the spatial dimensions of the high-level semantic features F_V^t and project them to the same channel dimension. The processed low-level features are then summed with F_V^t to obtain a combined feature representation. In the **concatenation fusion** approach, the resized multi-scale low-level features $F_{C_i}^t$ are concatenated with the high-level semantic features F_V^t along the channel dimension. A subsequent convolution layer projects the concatenated features to match the spatial dimensions of F_V^t , resulting in the fused feature. For **transformer decoder-based fusion**, we follow the common practice adopted in Mask2Former [50], utilizing high-level semantic features as queries and multi-scale low-level features as keys and values. (2) **Results:** The results in Table 7 demonstrate that our proposed feature injector outperforms other alternative fusion strategies across all datasets. While the transformer decoder-based fusion achieves competitive accuracy, the addition and concatenation methods are less effective at integrating multi-scale fine-grained details into the semantic features. In the last experiment, F_V serves as query, and F_C serves as key and value, yielding the best

Table 6

Study the impact of different pre-trained weights of ViT on three datasets, respectively. All results are described as percentages (%).

Model	Backbone	Pretrain	Pre-trained Data	Training Strategy	LEVIR-CD			WHU-CD			CLCD		
					F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
DMINet [33]	ResNet18	–	ImageNet(1k)	Supervised	90.71	82.99	99.07	88.69	79.68	98.97	67.24	50.65	95.21
GASNet [1]	ResNet34	–	ImageNet(1k)	Supervised	90.52	83.48	99.07	91.75	84.76	99.34	63.84	46.89	94.01
AMTNet [6]	ResNet50	–	ImageNet(1k)	Supervised	90.76	83.08	98.96	92.27	85.64	99.32	75.10	60.13	96.45
ChangeViT-T	ViT(Tiny)	Random Init	–	–	91.58	84.47	99.15	93.78	88.29	99.51	76.91	62.49	96.66
		DeiT-T [25]	ImageNet(1k)	Supervised	91.81	84.86	99.17	94.53	89.63	99.57	77.31	63.01	96.67
		Random Init	–	–	90.82	83.19	99.09	93.65	88.06	99.50	75.05	60.06	96.59
ChangeViT-S	ViT(Small)	DeiT-S [25]	ImageNet(1k)	Supervised	91.78	84.81	99.17	94.73	89.99	99.58	77.24	62.69	96.68
		DINO-S [26]	ImageNet(w/o labels)	Self-supervised	91.68	84.64	99.16	94.70	89.94	99.58	77.05	62.67	96.65
		DINOv2-S [27]	ImageNet(1k, 22k) & Google Landmarks	Self-supervised	91.98	85.16	99.19	94.84	90.18	99.59	77.53	63.30	96.76

Table 7

Study the impact of different modeling approaches for feature fusion on three datasets. Abbreviations Q, K, and V refer to query, key, and value, respectively.

Feature Fusion	Strategy	LEVIR-CD			WHU-CD			CLCD		
		F1	IoU	OA	F1	IoU	OA	F1	IoU	OA
Existing Approach	Addition	91.34	83.74	98.82	93.89	88.97	99.12	75.49	60.59	96.15
	Concatenation	91.22	84.31	99.01	94.30	89.15	99.28	75.65	60.91	96.32
	Transformer Decoder	91.54	84.65	99.12	94.39	89.63	99.45	75.82	60.98	96.45
Feature Injector (Proposed)	$F_V(K \& V), F_C(Q)$	91.78	84.80	99.17	94.60	89.75	99.58	75.84	61.08	96.58
	$F_V(Q), F_C(K \& V)$	91.98	85.16	99.19	94.84	90.18	99.59	77.53	63.30	96.76

Table 8

Performance comparison of different SCD methods on SECOND, FUSU, and HRSCD datasets, respectively. † denotes the results with our implementation. All results are described as percentages (%).

Method	SECOND				FUSU				HRSCD			
	F1	mIoU	OA	SeK	F1	mIoU	OA	SeK	F1	mIoU	OA	SeK
HRSCD-S4 [51]	58.21	71.15	86.62	18.80	55.13	66.77	24.26	66.94	69.39	67.79	81.32	22.50
SCDNet [52]	60.01	70.97	<u>87.40</u>	19.73	57.28	67.82	26.18	68.69	70.80	67.43	81.46	23.61
SSCD-1 † [53]	61.22	72.60	87.19	21.86	58.34	67.77	26.87	69.60	69.69	66.21	81.05	21.88
Bi-SRNet † [53]	61.85	72.08	87.20	21.36	58.65	68.49	27.75	69.69	71.72	67.83	<u>82.06</u>	24.59
DEFO † [54]	61.18	72.39	87.01	21.08	59.22	68.57	27.98	70.65	66.49	65.67	81.36	19.20
FUSU-Net † [55]	58.90	70.81	86.26	18.99	58.86	68.42	27.85	69.68	67.43	66.36	80.61	20.10
ChangeViT-T	<u>61.96</u>	<u>72.85</u>	86.78	<u>22.56</u>	<u>60.22</u>	<u>69.21</u>	<u>29.20</u>	<u>71.05</u>	<u>71.86</u>	<u>67.86</u>	81.89	<u>24.83</u>
ChangeViT-S	62.12	72.88	87.43	22.67	60.92	69.36	29.45	71.61	72.09	67.90	82.07	24.95

performance. This result is consistent with the conjecture mentioned in Section 3.2, suggesting that the feature injector effectively captures low-level value information most relevant to the high-level query and reintegrates it back to the query.

Effectiveness of ChangeViT on semantic change detection. To further investigate the effectiveness of ChangeViT on the semantic change detection (SCD) task, we conduct extensive experiments on three well-recognized SCD datasets: SECOND [56], FUSU [55], and HRSCD [51]. The experimental setup aligns with that of [2,55] to ensure a fair comparison. As presented in Table 8, ChangeViT achieves superior or competitive performance compared to existing methods across all datasets and evaluation metrics. This demonstrates ChangeViT's effectiveness in SCD tasks, regardless of the ViT model size (tiny or small). Notably, in challenging scenarios like the FUSU dataset, which involves a greater number of land cover classes and diverse changing objects, ChangeViT maintains superior performance. This further highlights ChangeViT's capability to handle both large-scale changes and capture fine-grained details effectively.

Size of changes v.s. performance. As depicted in Fig. 6(a), we conduct experiments on three datasets using the detail-capture module, ViT-S, and our proposed method to quantitatively analyse the performance of each method under different change sizes. The detail-capture module and ViT-S both integrate with a decoder which is the same as ChangeViT. The results indicate that the detail-capture module excels at detecting smaller changed targets, while the ViT-S demonstrates superiority in detecting larger ones. Our method capitalizes on ViT's powerful feature

expression while leveraging a detail-capture module for fine-detail information mining. This comprehensive approach enables superior performance across targets of all sizes.

Attention visualization. Fig. 7 illustrates the attention maps generated by the detail-capture module, plain ViT, and feature injector modules using two randomly selected images from the WHU-CD dataset. Two significant observations emerge from the attention maps: (1) Low-level features extracted by the detail-capture module primarily focus on edge and contour information, while being insufficient in perceiving large-scale objects, such as buildings. High-level semantic features effectively concentrate on buildings, but often result in blurred boundaries. (2) When integrated with the feature injector, the detailed information is effectively merged into the high-level semantic representation, leading to clearer and more complete boundaries. This demonstrates the collaborative synergy between the proposed two modules.

Qualitative comparison. We present representative visualization results on three datasets, comparing the performance of the detail-capture module, ViT-S, and our proposed method to demonstrate the effectiveness of ChangeViT. As shown in Fig. 6(b), the first row in each dataset presents the test results for smaller targets, while the second row corresponds to larger targets. From the results, we can see that the detail-capture module excels at detecting smaller targets, whereas ViT-S demonstrates superiority in detecting larger targets. The fundamental distinction lies in the local receptive field of CNN, enabling them to extract intricate local features, while ViT possesses a global receptive

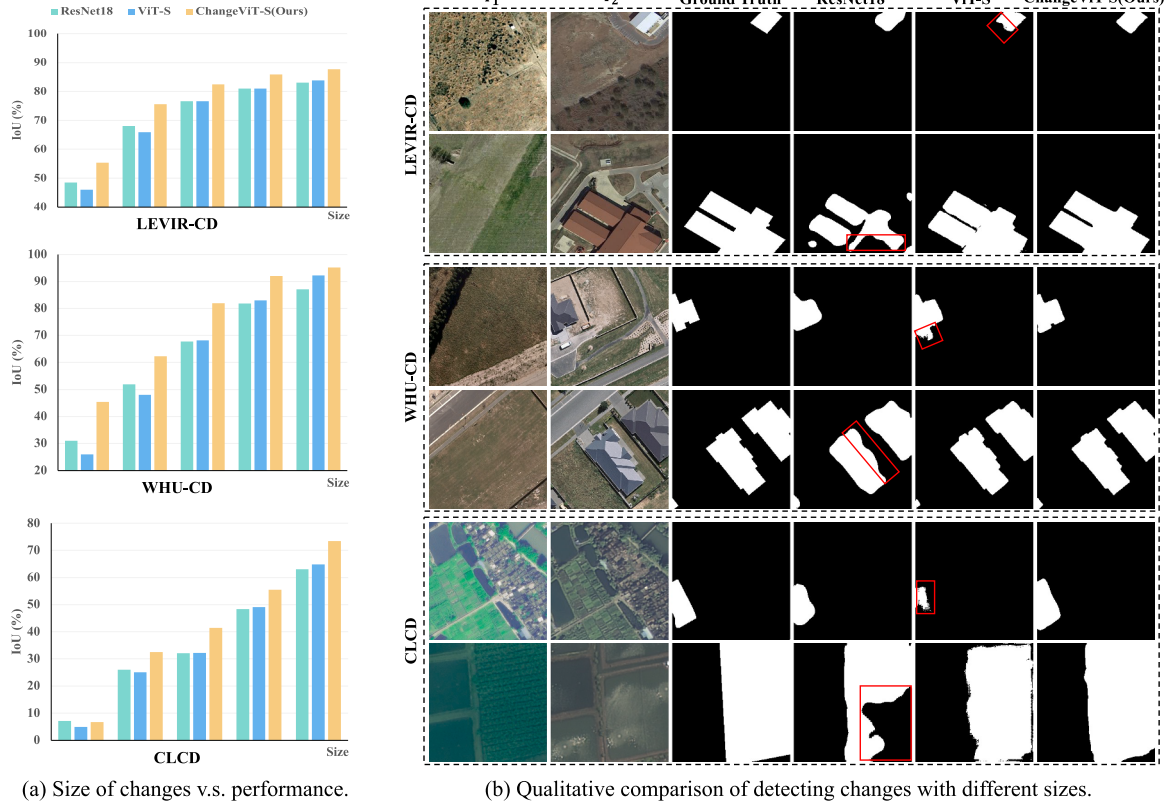


Fig. 6. (a) Each dataset is split into five intervals on average based on the change sizes. The horizontal axis incrementally reflects the change sizes, progressing from smaller to larger changes. (b) The predicted map within the red box indicates a poor detection outcome. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

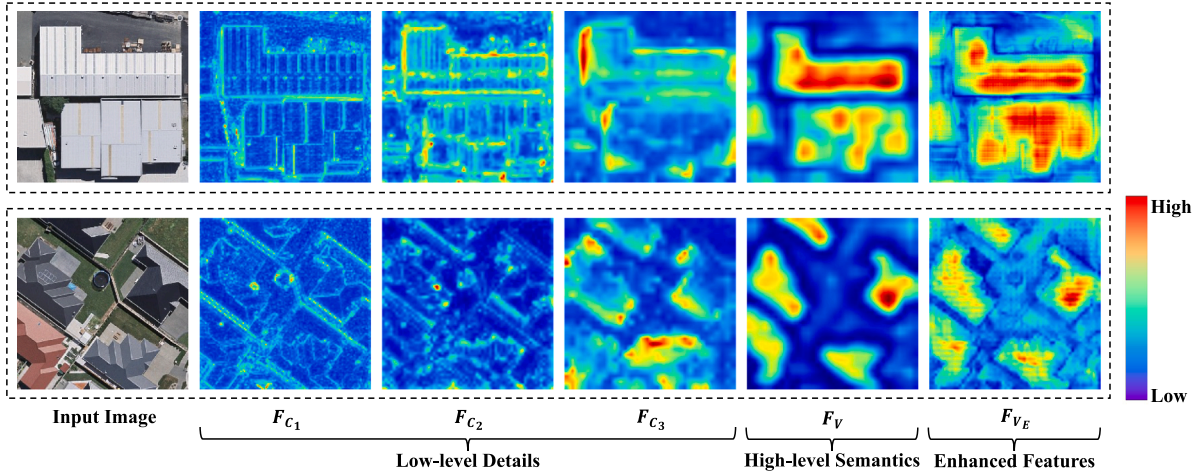


Fig. 7. Visualization of low-level detailed features (F_{C_1} , F_{C_2} , F_{C_3}) from the detail-capture module and high-level semantic features (F_V) from plain ViT. These are fused by the feature injector to produce an enhanced representation (F_{V_E}), effectively integrating details into semantic features. The color bar indicates the attention distribution.. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

field, facilitating the extraction of comprehensive global information. The proposed method efficiently integrates global and local information, resulting in superior performance.

To qualitatively compare with previous approaches, we present comprehensive samples covering small, large, sparse, and dense targets, as shown in Fig. 8. Yellow dashed rectangles highlight regions where ChangeViT surpasses other methods. Several key observations emerge: (1) ChangeViT consistently outperforms all compared methods across

various change scenarios, including **small** (samples 1, 2, 5, 10), **large** (samples 8, 11, 12), **sparse** (samples 1, 2, 9), and **dense** (samples 3, 4, 6, 7) targets. This superiority stems from the robust global modeling of ViTs and the detail-capture module's ability to extract intricate spatial features. The feature injector further integrates low-level spatial details into ViT's high-level semantics, enhancing ChangeViT's detection of diverse changes. (2) For **small** objects, ChangeViT captures subtle alterations with clearer boundaries, enabling precise identification of small-

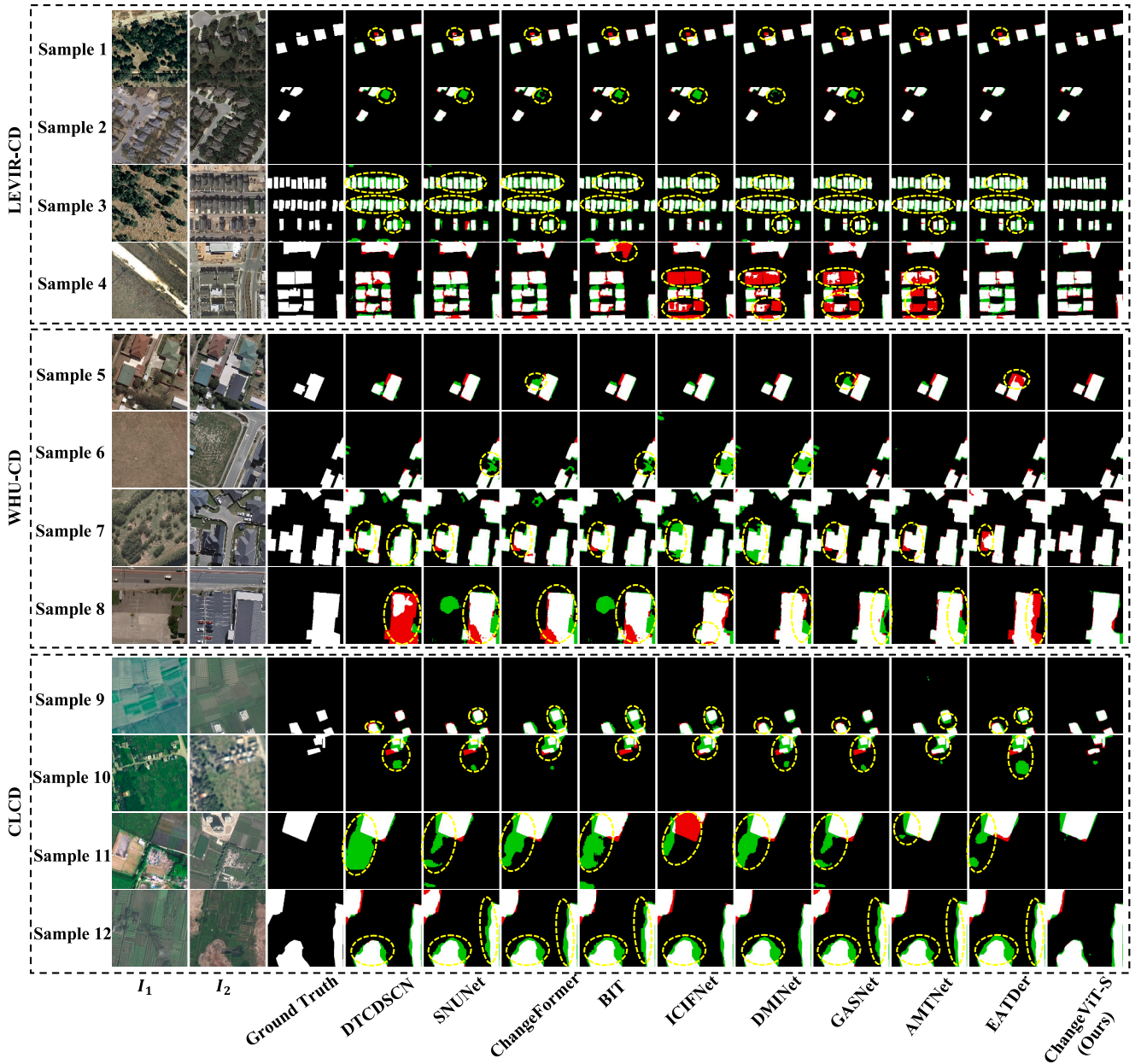


Fig. 8. Qualitative comparison of different methods on the three datasets. White represents a true positive, black is a true negative, green indicates a false positive, and red is a false negative. Fewer green and red pixels represent better performance. The area enclosed by the yellow dashed rectangle highlights the detection results within the regions of interest. For better clarity, please zoom in on the figure. (For interpretation of the references to colour in this figure legend, the reader is referred to the web version of this article.)

scale changes that other methods may overlook. (3) For **large** objects, ChangeViT yields more consistent semantic representations and accurate boundaries. (4) In **sparse** cases, it effectively detects isolated alterations without being affected by noise. (5) For **dense** scenes, ChangeViT maintains high accuracy and consistency, handling numerous changes in crowded environments. Overall, ChangeViT demonstrates versatility and superior performance across various change detection scenarios compared to existing methods.

5. Limitation and discussion

While ChangeViT demonstrates state-of-the-art performance on four widely recognized change detection benchmarks, several limitations remain. First, the potential of ChangeViT in advanced differential feature modeling and decoder design has not been fully explored, which may

further enhance its capability for complex change scenarios. In addition, although ChangeViT effectively integrates fine-grained spatial information through its detail-capture module and feature injector, its application is currently limited to change detection tasks. Future work could extend ChangeViT to related tasks such as change captioning [57,58], image fusion [59], or other multi-modal remote sensing applications. These directions are left for future research.

6. Conclusion

In this paper, we present ChangeViT, a simple yet effective framework that leverages a plain Vision Transformer (ViT) as its primary feature extractor for large-scale change detection in remote sensing imagery. Enhanced by a detail-capture module and a feature injector, ChangeViT seamlessly integrates fine-grained spatial details with high-

level semantic representations. Extensive experiments on four benchmark datasets validate that ChangeViT outperforms existing hierarchical models across all evaluation metrics, revealing the untapped potential of vanilla ViTs for change detection. Comprehensive analyses and visualizations further demonstrate the effectiveness of each module. We hope this work provides valuable insights and inspires further research into applying plain ViT architectures for broader computer vision tasks.

CRedit authorship contribution statement

Duowang Zhu: Writing – review & editing, Writing – original draft, Methodology, Investigation; **Xiaohu Huang:** Writing – review & editing, Methodology, Conceptualization; **Haiyan Huang:** Writing – review & editing, Visualization, Investigation, Conceptualization; **Qimin Cheng:** Writing – review & editing, Supervision, Formal analysis; **Zheng Huang:** Visualization, Validation; **Zhenfeng Shao:** Writing – review & editing, Supervision, Methodology, Funding acquisition.

Data availability

Data will be made available on request.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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